

Factoring 7531 using the Continued Fractions Method

Find a non-trivial factor of $N = 7531$ using the continued fractions method.

1) Continued fraction of $\sqrt{7531}$

First, $a_0 = \lfloor \sqrt{7531} \rfloor = 86$ because $86^2 = 7396$ and $87^2 = 7569$.

We use the standard recurrence for the continued fraction of $\sqrt{N}$:

- Initialize: $m_0 = 0, d_0 = 1, a_0 = \lfloor \sqrt{N} \rfloor$.
- For $k \geq 0$: $m_{k+1} = d_k a_k - m_k, \quad d_{k+1} = \frac{N - m_{k+1}^2}{d_k}, \quad a_{k+1} = \left\lfloor \frac{a_0 + m_{k+1}}{d_{k+1}} \right\rfloor$.

The first few triplets (m_k, d_k, a_k) are:

k	m_k	d_k	a_k
0	0	1	86
1	86	135	1
2	49	38	3
3	65	87	1
4	22	81	1
5	59	50	2

So the continued fraction begins: $\sqrt{7531} = [86; 1, 3, 1, 1, 2, \dots]$

2) Convergents and the r_k values

Convergents are computed from partial quotients a_k using:

- $p_{-2} = 0, p_{-1} = 1; q_{-2} = 1, q_{-1} = 0$
- $p_k = a_k p_{k-1} + p_{k-2}, \quad q_k = a_k q_{k-1} + q_{k-2}$.

Then compute $r_k = p_k^2 - N q_k^2$.

k	a_k	p_k	q_k	$r_k = p_k^2 - 7531 q_k^2$
0	86	86	1	-135
1	1	87	1	38
2	3	347	4	-87
3	1	434	5	81

2	3	347	4	-87
3	1	434	5	81
4	1	781	9	-50
5	2	1996	23	117

At $k = 3$ we get a perfect square immediately:

$$p_3 = 434, \quad q_3 = 5$$

$$r_3 = 434^2 - 7531 \cdot 5^2 = 188356 - 188275 = 81 = 9^2.$$

3) Congruence of squares and extracting a factor

From $r_3 = p_3^2 - Nq_3^2$ we rearrange:

$$p_3^2 - 9^2 = N \cdot q_3^2 = 7531 \cdot 25.$$

Therefore, $434^2 \equiv 9^2 \pmod{7531}$.

So 7531 divides $(434 - 9)(434 + 9)$:

$$(434 - 9)(434 + 9) = 425 \cdot 443.$$

Now compute gcds:

$$\text{I. } \gcd(434 - 9, 7531) = \gcd(425, 7531)$$

Euclid steps:

step	a	b	$a = b \cdot q + r$
1	7531	425	$7531 = 425 \cdot 17 + 306$
2	425	306	$425 = 306 \cdot 1 + 119$
3	306	119	$306 = 119 \cdot 2 + 68$
4	119	68	$119 = 68 \cdot 1 + 51$
5	68	51	$68 = 51 \cdot 1 + 17$
6	51	17	$51 = 17 \cdot 3 + 0$

Hence, $\gcd(425, 7531) = 17$.

$$\text{II. } \gcd(434 + 9, 7531) = \gcd(443, 7531)$$

Since $7531 = 443 \cdot 17$, we have $\gcd(443, 7531) = 443$.

Result

We found non-trivial factors of 7531:

$$7531 = 17 \cdot 443.$$